Department of Computer Science

Faculty of Computing

Assignment 1 : Business Insight

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1.0 INTRODUCTION

The dataset used in this analysis is derived from the Behavioral Risk Factor Surveillance System (BRFSS) 2015 survey, focusing on diabetes and various health indicators. It consists of 253,680 entries and 22 numerical features, all recorded in a consistent and clean format without any missing values. Each feature represents an important health, lifestyle or demographic factor, including conditions such as high blood pressure, cholesterol levels, BMI, smoking habits, physical activity, dietary habits, alcohol consumption, healthcare access and general health status. The dataset categorizes individuals into three classes which is no diabetes (0), pre-diabetes (1) and diabetes (2). The dataset captures a wide demographic range, providing a comprehensive foundation for health analytics, risk prediction and healthcare planning. The clean structure and large sample size allow for meaningful insights and statistically reliable conclusions.

2.0 BUSINESS INSIGHTS

The following section presents key business insights derived from the analysis of the BRFSS 2015 health indicators dataset. Each insight highlights important trends and relationships within the data that are relevant for healthcare planning, public health strategy and resource allocation. By understanding patterns related to diabetes prevalence, lifestyle factors, healthcare accessibility and demographic influences, organizations can make more informed decisions to improve population health outcomes. Each insight is accompanied by an elaboration to explain its significance and potential applications.

2.1 Prevalence of Diabetes in the Population

The analysis of the BRFSS 2015 dataset reveals that approximately 29.7% of the surveyed population is diagnosed with either pre-diabetes or diabetes. This high prevalence highlights a major public health concern, suggesting that nearly one in three individuals faces an increased risk of developing severe health complications if timely interventions are not implemented. Recognizing the scale of the issue allows healthcare providers to prioritize resources for early detection programs, education campaigns and accessible treatment plans aimed at diabetes prevention and management.

Early intervention is crucial, as it can significantly delay or prevent the onset of type 2 diabetes and reduce associated healthcare costs. Public awareness initiatives focusing on lifestyle changes, such as healthier eating, increased physical activity and smoking cessation can have a major impact in reducing new diabetes cases. Moreover, healthcare systems must ensure that adequate medical support, screening facilities and affordable options are readily available, especially for high-risk groups.

In conclusion, the insight into diabetes prevalence provides essential guidance for policymakers, healthcare planners and community organizations to develop strategies that not only treat diabetes but also prevent its spread, ultimately improving the overall health and well-being of the population.

2.2 Impact of Lifestyle Factors on Health Outcomes

Lifestyle behaviors such as smoking, physical activity and diet play a critical role in determining health outcomes. Analysis of the dataset reveals that 44.3% of the respondents identify as smokers, while 75.6% report participating in regular physical activity and 63.4% consume fruits daily. These figures suggest that although a majority engage in positive behaviors like exercise and healthy eating, a significant portion still participates in high-risk habits, particularly smoking.

Understanding these patterns is crucial for designing effective public health strategies. Encouraging healthier behaviors through targeted campaigns, such as smoking cessation programs, community fitness initiatives and nutrition education that can lead to a substantial decrease in the prevalence of chronic diseases like diabetes, heart disease and stroke.

Promoting a balanced approach that includes physical activity and healthy eating while reducing smoking rates can greatly enhance long-term population health. These insights provide policymakers and healthcare providers with the foundation to implement lifestyle which focused interventions that not only treat existing conditions but also prevent future health complications.

2.3 Cholesterol Check Behavior and Early Detection

The variable CholCheck shows whether a person has had a cholesterol check in the past five years. Those who skip routine checks are at greater risk of undetected diabetes and related complications.

The business implication is that encouraging regular health screenings, such as cholesterol tests, can lead to earlier detection of diabetes, which organizations can support through wellness programs and public health campaigns

2.4 Key Risk Factors for Diabetes:

General Health: Individuals with poorer general health (GenHlth) have a significantly higher risk of diabetes. This highlights the importance of overall well-being in diabetes prevention.

High Blood Pressure and High Cholesterol: HighBP and HighChol are strongly correlated with diabetes, indicating that managing these conditions is crucial for mitigating diabetes risk.

BMI: Higher BMI values are associated with increased diabetes prevalence. This emphasizes the need for weight management strategies in diabetes prevention and control.

Age: Diabetes prevalence increases with age, suggesting the need for targeted interventions and screening programs for older populations.

Income and Education: Lower income and education levels are linked to higher diabetes rates, highlighting the social determinants of health and the need for accessible healthcare resources.

2.5 Gender Differences in Diabetes

The analysis of the BRFSS 2015 dataset shows that males may have a little greater rate of diabetes compared to females. Several factors may contribute to this difference, including biological variations, occupational exposures, lifestyle behaviors, and differences in healthcare-seeking patterns.

To create efficient and focused intervention methods, it is essential to acknowledge the gender disparities in disease frequency. The impact of public health programs can be significantly increased by health initiatives that consider gender-specific risks. For instance, wellness programs in male-dominated workplaces could prioritize regular health screenings, promote physical activity, and encourage healthier eating habits. Furthermore, public campaigns aimed at reducing the stigma surrounding men's engagement with preventive healthcare could lead to earlier detection and better management of diabetes among men.

Understanding and addressing gender differences in diabetes allow healthcare providers, policymaker and employers to design more precise, targeted interventions that increase engagement, encourage healthier behaviors and ultimately reduce the rate of diabetes across both genders. Gender consideration not only promote health fairness but also guarantee more efficient use of healthcare resources.

2.6 Mobility Limitation and Diabetes Risk

The analysis of the BRFSS 2015 dataset highlights a strong correlate between mobility limitation and the rate of diabetes. Diabetes and prediabetes are substantially more common in people who report difficulty walk (DiffWalk) than in people who do not have mobility problem. Limited physical function can both result from and contribute to diabetes progression, creating a continuous loop of poor health.

To address this, healthcare providers should implement low-impact exercise programs designed for individual with mobility issues. Public health planner can invest in accessible fitness solution to ensure that all individuals, regardless of physical limitations, have opportunities to maintain an active lifestyle.

Understanding the connection between mobility and diabetes is crucial to designing inclusive and effective intervention strategies. By promoting physical activity among those with limitations, we can reduce disease progression, improve quality of life and lower long-term healthcare costs, like insulin cost such as the expenses associated with insulin therapy and diabetes-related complications.

3.0 Dataset Overview and Preparation for Analysis

3.1 Dataset Overview:

The dataset contains information on various health indicators and demographic variables collected from individuals through the Behavioral Risk Factor Surveillance System (BRFSS) in 2015. It includes features such as age, sex, BMI, blood pressure, cholesterol levels, general health status, and diabetes status (Diabetes_012). The target variable Diabetes_012 represents the presence or absence of diabetes (0: No diabetes, 1: Prediabetes or borderline diabetes, 2: Diabetes).

3.2 Numerical and Categorical Features:

The dataset contains a mix of numerical (e.g., age, BMI) and categorical (e.g., sex, education level) features. Data Distribution: The distributions of numerical features vary. Some features, like BMI, may exhibit skewness or have outliers, requiring potential transformations during analysis. Missing Values: The dataset does not contain any missing values, which simplifies the data cleaning process.

Correlations: There are significant correlations between certain health indicators and diabetes status. Notably, GenHlth, HighBP, and BMI exhibit strong correlations with Diabetes_012.

Feature Interactions: The analysis reveals potential interactions between features, suggesting that combinations of health indicators can influence diabetes risk.

3.3 Insights from Analysis:

The analysis identifies several key risk factors for diabetes, including poor general health, high blood pressure, high cholesterol, high BMI, older age, lower income, and lower education levels. Potential Confounding Factors: The relationships between certain features may act as confounding factors, influencing the observed correlations with diabetes. Predictive Modeling: Classification models trained on the dataset demonstrate the potential for predicting diabetes risk using the available health indicators.

3.4 Future Analysis

To strengthen the findings and improve the predictive models, several areas can be explored in future analyses:

- **Investigating Confounding Factors:** Some variables may influence both diabetes and other health outcomes, which can distort the results. Techniques such as stratification or regression analysis could help uncover these hidden influences and improve model accuracy.
- **Feature Engineering:** New features can be created by combining existing variables, such as age and BMI, or by introducing interaction terms. This process may help the model capture more complex relationships within the data.
- **Model Optimization:** The performance of the prediction models can be further improved by tuning hyperparameters or trying out alternative algorithms. This could lead to better classification accuracy and more reliable risk assessments.
- **Enhancing Model Interpretability:** Making models easier to interpret is important, especially for use in healthcare settings. By focusing on models that clearly show how decisions are made, stakeholders can better understand the key factors contributing to diabetes risk.

3.5. Integration with External Data:

Integrating this dataset with other relevant data sources (e.g., genetic information, lifestyle data) could provide a more comprehensive understanding of diabetes risk and potentially improve prediction accuracy.

4.0 CONCLUSION

The diabetes health indicators dataset helps us explore how different health factors are related to the risk of developing diabetes. The insights from this data can guide prevention efforts, help with planning healthcare resources, and support more personalized care. To get even more useful results, we can dig deeper into how different factors interact, improve how the data is used, and refine prediction models. Adding in outside data could also give a fuller picture and lead to better understanding.

5.0 APPENDIX

Dataset : https://drive.google.com/drive/u/1/folders/1js3-Z6LdT-dQf0f7IJVqMOYBUSH4Iuz5